Department of Mechanical Engineering
COEP Technological University
Pune, India
nkc.mech@coeptech.ac.in

November 3, 2026

The Editor-in-Chief
Reliability Engineering & System Safety

Dear Editor,

We submit the manuscript "Evidence, not verdicts: what 590 public PX4 crash logs can and cannot tell a drone fleet operator" as a research article.

Flight logs are increasingly used to decide why a small drone crashed, who was responsible, and which aircraft to inspect next. We tested those uses on the largest public archive of real autopilot logs (PX4 Flight Review, 479,547 logs). Each confirmatory test was committed to version control with its pass mark before it was run; analyses added during review are marked as exploratory.

Two findings are, to our knowledge, new and apply to any evaluation of log-based diagnostics:

- **Evaluation choices shape the apparent signal.** In a factorial audit of 32 evaluation settings, a crash-vs-normal AUC of 0.88 fell by 0.21 [0.06, 0.34]; the one choice that could be isolated, grouping cross-validation folds by vehicle, accounted for 0.06. Comparing each drone with its own earlier flights brought it to chance.
- **Archive records need care.** The usual tilt rule finds no crash-like tilt in 61% of crash-rated multicopter logs and misses 37–58% of crashes, depending on the reviewer; pilot-rated logs show one less often (23% vs 45%); and dates cannot order flights within a day.

These rest on declared tests:

- neither a declared rule nor cross-validated models separated pilot-caused from aircraft-caused crashes (balanced accuracy 0.47 on 112 held-out crashes), nor recovered a blind relabelling;
- crash-window leakage was small overall but material among multicopter crashes with a crash-like tilt;
- the flight that crashed was no different from the same drone's earlier flights, whereas drones that later crashed vibrated more than matched drones on ordinary flights, a suggestive signal ($p = 0.012$ per airframe, 0.076 per operator site) that did not clearly beat random inspection against 412 non-crash drones;
- a published trained fault detector, re-implemented, fell from AUC 0.94 to 0.63 once whole days were held out, and separated fault flights before any fault existed (AUC 0.85).

The findings feed a reporting checklist and an open-source tool that presents evidence rather than verdicts; code, declarations, derived features and labels are released.

The manuscript is original, has not been published and is not under consideration elsewhere. All authors approve the submission. An AI assistant was used in preparing code and text, as declared in the manuscript.

Yours sincerely,

Dr. Nagesh Chougule
on behalf of all authors